THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assignment 3, 2025

School of Mathematics and Statistics

MAST20009 Vector Calculus

Submission deadline: 5pm on Monday May 5

This assignment consists of 10 pages (including this page) with 3 questions and 37 total marks

Instructions to Students

- This assignment is worth 5% of your total MAST20009 mark.
- If you have a printer, print the assignment one-sided.

Writing

- Assignments must be neatly written in blue or black pen or may be typeset using LaTeX.
- Full working must be shown in your solutions.
- Marks will be deducted for incomplete working, insufficient justification of steps, incorrect mathematical notation and for messy presentation of solutions.
- If you are writing answers on the assignment and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned assignment as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary.

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Question 1 (10 marks)

Let $\mathbf{F}$ be the vector field in $\mathbb{R}^3$ given by

$$\mathbf{F}(x, y, z) = (xz + xy)\,\mathbf{i} + \alpha(yz - xy)\,\mathbf{j} + \beta(yz + xz)\,\mathbf{k}$$

where $\alpha, \beta \in \mathbb{R}$ are constants. Suppose that $\mathbf{F}$ satisfies $\mathbf{F} = \nabla \times \mathbf{G}$ for some smooth vector field $\mathbf{G}$.

(a) Find α and β.

(b) Now suppose that

$$\mathbf{G}(x, y, z) = xyz\,\mathbf{i} - xyz\,\mathbf{j} + g(x, y, z)\,\mathbf{k}$$

for some function g. Find $g(x, y, z)$.

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Question 2 (16 marks)

(a) Consider the curve C made up of two straight line segments, from $P = (0, 0, 0)$ to $Q = (1, 2, 2)$, and then from Q to $R = (4, 2, -2)$. Compute the integral

$$\int_C (xz + y)\, ds.$$

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(b) Let C be the circle of radius 2 with centre $(1,1)$, and let C_1 be the curve along one quarter of C, from $(3,1)$ to $(1,3)$ in the anticlockwise direction. Define the force field $\mathbf{F}(x,y) = (2x, x-y)$. Find the work done by $\mathbf{F}$ in moving a particle along C_1.

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Question 3 (11 marks)

Let D be the region in $\mathbb{R}^2$ between the parabola $y = x^2$ and the line $y = 2 - x$. Let $f(x, y) = \frac{x}{y}$.

(a) Consider the integral

$$I = \iint_D f(x, y)\, dA.$$

Write down an expression for I as a double integral with *vertical* strips. You should include a sketch of the region D, labelling the relevant features. You *do not* need to evaluate the integral.

(b) Now compute the integral I, this time using *horizontal* strips.

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End of Assignment — Total Available Marks = 37